

Deductive, Inductive & Abductive Reasoning

The direction of inference between theory and data

PROFESSIONAL DEFINITION

Reasoning describes how a study moves between theory and data. Deduction starts from existing theory, derives hypotheses and tests them against data (theory to data). Induction starts from observations and builds toward theory (data to theory). Abduction moves back and forth, starting from a surprising observation and working out the most plausible explanation. The choice signals how your study generates knowledge and must match your philosophy (Saunders, Lewis and Thornhill, 2019).

DEDUCTIVE (theory-first)

Begin with a theory or hypothesis, collect data to test it, confirm or reject. Associated with positivism and quantitative designs. Strength: rigorous, replicable. Risk: can miss what the pre-set frame does not anticipate.

INDUCTIVE (data-first)

Begin with data, identify patterns, build explanation or theory. Associated with interpretivism and qualitative designs. Strength: openness to the unexpected. Risk: harder to generalise; demands careful, transparent analysis.

Abduction — the pragmatic middle. Most real business research is abductive in practice: you notice something puzzling, propose a candidate explanation, and iterate between data and theory to refine it. Naming abduction honestly is often more accurate than forcing your study into a pure deductive or inductive box.

Feature	Deductive	Inductive	Abductive
Starts from	Theory	Data	A puzzle
Direction	Theory to data	Data to theory	Back and forth
Aim	Test	Build	Best explanation
Fits	Positivism	Interpretivism	Pragmatism

MATCH THE CHAIN

Reasoning must sit consistently with your philosophy and design. Deductive-interpretivist or inductive-experimental pairings raise flags.

STATE HYPOTHESES OR NOT

Deductive studies state testable hypotheses up front. Inductive studies do not — don't bolt hypotheses onto exploratory qualitative work.

BE HONEST

If you iterated between data and theory, say abductive. Claiming pure induction while using a coding framework is a contradiction.

Common pitfalls. (1) Claiming a deductive study but having no hypotheses. (2) Claiming induction while coding to a pre-existing model. (3) Not naming the reasoning at all. (4) A reasoning type that contradicts the stated philosophy.

REFERENCES

Saunders, M., Lewis, P. and Thornhill, A. (2019) *Research Methods for Business Students*. 8th edn. Harlow: Pearson. · Bryman, A. and Bell, E. (2015) *Business Research Methods*. 4th edn. Oxford: Oxford University Press.